

Bit-Plane A

DP8 (8×4) — decomposition over the bit planes of A

Step 1 — the dot product

$$\sum_{i=0}^7 A_i B_i$$

Step 2 — expand A into its bit planes and exchange the sums

Each A_i is an 8-bit value, $A_i = \sum_{j=0}^7 2^j A_i^j$, so

$$\sum_{i=0}^7 A_i B_i = \sum_{i=0}^7 \left[\sum_{j=0}^7 2^j A_i^j \right] B_i = \sum_{j=0}^7 2^j \sum_{i=0}^7 A_i^j B_i$$

$A_i^j B_i$ is a selection, not a product.

Step 3 — pair the lanes

$$\sum_{i=0}^7 A_i B_i = \sum_{j=0}^7 2^j \sum_{i=0}^3 \underbrace{\left[A_{2i}^j B_{2i} + A_{2i+1}^j B_{2i+1} \right]}_{R(A_{2i}^j, A_{2i+1}^j)}$$

$R(A_{2i}^j, A_{2i+1}^j)$ depends on the two bits A_{2i}^j, A_{2i+1}^j and takes one of four values:

A_{2i}^j	A_{2i+1}^j	$R(A_{2i}^j, A_{2i+1}^j)$
0	0	0
0	1	B_{2i+1}
1	0	B_{2i}
1	1	$B_{2i} + B_{2i+1}$

Step 4 — split the work

Every entry of the table is a function of B alone, so the table is built outside the PE and only the selection stays inside:

$$\underbrace{\{0, B_{2i}, B_{2i+1}, B_{2i} + B_{2i+1}\}}_{\text{dispatcher}} \longrightarrow \underbrace{\sum_{j=0}^7 2^j \sum_{i=0}^3 R(A_{2i}^j, A_{2i+1}^j)}_{\text{PE}}$$

A is the wide operand: 8 planes × 4 lane pairs = 32 selections of 6 bits per DP8, twice the rows of Bit-Plane B.